

Joshua C. Macdonald

Computational Scientist

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Research Profile

Computational scientist whose work determines what actions to take, what experiments to run, and what measurements are worth collecting in systems where interventions are costly and uncertainty is unavoidable. I develop methods, scientific AI/ML, and research software for partially observed systems across earth, environmental, and health sciences — integrating generative modeling (mechanistic, statistical, and hybrid), Bayesian inference, uncertainty quantification, and scientific computing to build defensible decision support. Current application areas include infectious disease forecasting and intervention timing, surveillance design, marine and terrestrial ecology, and cultural transmission dynamics.

Education

PhD	University of Louisiana at Lafayette , Mathematics	Lafayette, Louisiana, USA
	• Advisor: Hayriye Gulbudak • Dissertation: <i>Population Dynamics Under Disturbance in Ecology and Epidemiology</i>	2017 – 2022
MS	University of Louisiana at Lafayette , Mathematics	Lafayette, Louisiana, USA
		2017 – 2020
BA	University of North Carolina Asheville , Mathematics	Asheville, North Carolina, USA
		2014 – 2016
BA	University of North Carolina Greensboro , Archaeology	Greensboro, North Carolina, USA
		2008 – 2013

Appointments

Johns Hopkins Bloomberg School of Public Health , Postdoctoral Scholar	Baltimore, Maryland, USA
International Vaccine Access Center, Department of International Health. Mentor: Shaun Truelove. Research focus: infectious disease modeling and inference, scientific computing, decision support systems, and research software engineering.	2024 – present
Tel Aviv University , Zuckerman STEM Leadership Postdoctoral Fellow	Tel Aviv, Israel
School of Zoology. Mentor: Yoav Ram. Research drew directly on archaeological training in the study of material culture diversity and cultural transmission, applying Bayesian dimensionality reduction and multivariate data analysis to cultural datasets.	2022 – 2024

Computational Expertise

Modeling: Generative models (ODE/PDE/stochastic/hybrid); compartmental and agent-based models; scientific AI/ML (physics-embedded surrogates, lawful learning); multivariate data analysis and dimensionality reduction; stability, bifurcation, and sensitivity analysis; parameter identifiability and inverse problems

Inference: Bayesian hierarchical models; variational and simulation-based inference; data assimilation; uncertainty quantification; model calibration and validation; proper scoring rules and forecast evaluation

Computing: Python, Julia, C++, Stan, R, MATLAB. Scientific computing; high-performance numerical solvers; operator splitting; data pipelines and exploratory analysis; reproducible workflows (Git, CI/CD, containerization)

Published Papers

- Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo** 2024
J.C. Macdonald, H. Gulbudak, B. Beechler, E. Gorsich, S. Gubbins, E. Perez, A. Jolles
[10.1086/730703](https://doi.org/10.1086/730703) (The American Naturalist)
- Cultural transmission, networks, and clusters among Austronesian-speaking peoples** 2024
J.C. Macdonald, J. Blanco-Portillo, M.W. Feldman, Y. Ram
[10.1017/ehs.2024.45](https://doi.org/10.1017/ehs.2024.45) (Evolutionary Human Sciences)
- Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms** 2023
J.C. Macdonald, H. Gulbudak
[10.1007/s00285-023-01969-7](https://doi.org/10.1007/s00285-023-01969-7) (Journal of Mathematical Biology)
- First and second wave outbreak size with reactive quarantine measures: a COVID-19 model applied to an epidemic in China** 2022
C.J. Browne, H. Gulbudak, J.C. Macdonald
[10.1016/j.jtbi.2021.110919](https://doi.org/10.1016/j.jtbi.2021.110919) (Journal of Theoretical Biology)
- Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates** 2021
J.C. Macdonald, C.J. Browne, H. Gulbudak
[10.1098/rsos.210227](https://doi.org/10.1098/rsos.210227) (Royal Society Open Science)

Papers Under Review

- An immuno-epidemiological model of foot-and-mouth disease in African buffalo with carrier transmission** 2026
S. Atkins, H. Gulbudak, S. Welker, H. Smith, *J.C. Macdonald*, B.R. Beechler, A.E. Jolles
(Under review, Bulletin of Mathematical Biology)

Preprints

- Bayesian estimation of the number of significant principal components for cultural data** 2024
J.C. Macdonald, J. Blanco-Portillo, M.W. Feldman, Y. Ram
[10.48550/arXiv.2409.12129](https://doi.org/10.48550/arXiv.2409.12129)

Manuscripts in Preparation

- Convergence characterization of coordinate-ascent variational inference for Bayesian PCA** 2026
J.C. Macdonald, Y. Ram
(In preparation)
- Posterior predictive eigenvalue testing for spectral dimensionality selection** 2026
J.C. Macdonald, S. Naim, Y. Ram
(In preparation for Bioinformatics or PLOS Computational Biology)
- Lawful Learning: Embedding Physical Laws into Surrogate Representations for Robust Extrapolation** 2026
J.C. Macdonald
(In preparation for Proceedings of the Royal Society A)

Structured observables: recovering dynamical structure from parameter-indexed generators J.C. Macdonald (In preparation for SIAM Journal on Applied Dynamical Systems)	2026
Shapelet analysis of infectious disease forecasts reveals heterogeneity across epidemic phases K. Roosa, L.A. White, T.M. Leon, J.C. Macdonald , A. Srivastava, S. Truelove (In preparation for Epidemics)	2026

Research Support

National One Health Mission India (\$1M / 3 years) — Pending Development of an Integrated One Health Disease Modelling and Digital Health Program in Human, Livestock, and Wildlife. Co-Investigator. Research role: mathematical model parameterization and data analysis; mechanistic modeling for HPAI and anthrax in India; stakeholder training in modeling; and supervision of postdoctoral and doctoral trainees.	In review
USDA-NIFA AFRI US–UK Collaboration (Award #1032897) Quiescence to emergence: Does nairovirus diversity loss increase disease transmission, severity and the risk of emergence? US PIs: Brianna Beechler, Anna Jolles, Cameron Browne, and Hayriye Gulbudak; linked UKRI award BB/Z516934/1 with Emma Thomson, Teresa Lambe, Roman Biek, and Georgina Limon-Vega. Collaborator (40% FTE). Research role: mathematical model parameterization, data analysis, and advising on mathematical and statistical methods.	2025 – 2028
USDA-ARS Cooperative Agreement (3022-32000-021-018-S) Investigating the Epidemiology of CCHFV in an Endemic Setting: Uganda. PIs: Bree Beechler and Anna Jolles (Oregon State University). Collaborator. Research role: mathematical modeling of cross-species CCHFV transmission dynamics, surveillance design optimization, and advising on mathematical and statistical methods for decision support.	2025 – present
CDC-RFA-FT-23-0069 (NU38FT000012) Atlantic Coast Center for Infectious Disease Dynamics and Analytics (ACCIDDA). Directors: Justin Lessler, Kimberly Powers, and Shaun Truelove. Postdoctoral Scholar (60% FTE). Insight Net center for infectious disease forecasting and outbreak analytics. Research role: operational modeling, forecasting pipeline development, and computational infrastructure.	2023 – present

Research Software – Lead Developer

OP System (Operator-Partitioned System) ACCIDDA. Declarative specification language and compiler for structured dynamical systems. Researchers define models via explicit equations or transition diagrams with multi-axis stratification; specifications compile to validated, safe bytecode closures with structured metadata passed to downstream solvers. Specification layer of the operator-partitioned simulation stack. github.com/ACCIDDA/op_system	2026 – present
pp-eigentest Posterior predictive eigenvalue testing for determining signal rank in high-dimensional datasets. Three-layer consensus architecture combining dimensionality heuristics, adaptive thresholding, and multiple testing correction (FWER, FDR). NumPy and JAX backends; designed to consume VBPCapy posterior outputs. github.com/yoavram-lab/pp-eigentest	2026 – present

Trade Study (Python + Julia)

2026 – present

Design and evaluation framework for scientific simulation studies. Simulators with known latent state provide controlled environments to score competing configurations — model formulations, solver choices, measurement strategies, or any design decision — against ground truth using proper scoring rules and multi-objective Pareto optimization. Hierarchical phases, Bayesian model stacking, and global sensitivity analysis. Available in Python and Julia.

- github.com/jcm-sci/trade-study (Python)
- github.com/jcm-sci/TradeStudy.jl (Julia)
- pypi.org/project/trade-study/
- [doi:10.5281/zenodo.19599838](https://doi.org/10.5281/zenodo.19599838)

OP Engine (Operator-Partitioned Engine)

2025 – present

ACCIDDA. Operator-partitioned ODE/PDE solver core. Splits the right-hand side into explicit and implicit parts, advancing them jointly via nine methods (explicit, IMEX, fully implicit) with adaptive step-size control. Cached implicit solves with dense/sparse autodispatch; zero per-step allocation. Consumes OP System specifications.

- github.com/ACCIDDA/op_engine

vbpcapy

2024 – present

Variational Bayesian PCA for incomplete data with native per-entry missingness handling, full posterior uncertainty quantification, automatic component pruning, and built-in model selection. Missing-aware preprocessing pipeline and C++-accelerated kernels; scikit-learn-compatible API. Applied to genetic, cultural, and ecological datasets.

- pypi.org/project/vbpcapy/
- [doi:10.5281/zenodo.19389250](https://doi.org/10.5281/zenodo.19389250)

Research Software – Collaborative Infrastructure

FlepiMoP2 (Flexible Epidemic Modeling Pipeline 2)

2025 – present

ACCIDDA. Contributor to the configuration-driven orchestration engine for CDC-supported infectious disease forecasting and scenario analysis. Plugin architecture decouples model specification, numerical integration, and output persistence.

- [Documentation](#)

TabularEpi

2025 – 2025

UNC / ACCIDDA. Tabular epidemic simulator supporting mixed deterministic-stochastic models. Conducted systematic performance benchmarking that identified bottlenecks, informing the design of the next-generation pipeline.

FlepiMoP (Flexible Epidemic Modeling Pipeline)

2024 – 2025

Hopkins IDD / ACCIDDA. Developed simulation and statistical inference components for operational influenza modeling; conducted benchmarking that identified 5–20x performance improvements, informing the next-generation pipeline architecture.

Research Software – Operational Deployments

Hib Vaccination Modeling (Navajo Nation)

2026 – present

ACCIDDA. Technical supervisor managing PhD students in implementation of an age- and immune-status-structured Hib model for the Navajo Nation to evaluate the impact of long-running vaccination programs.

Flu Scenario Modeling Hub ACCIDDA. Lead Model Developer across the 2024/25, 2025/26, and 2026/27 seasons. Set up the ACCIDDA flu model and operational pipeline (2024/25). Extensive work with external modeling packages drove contributions that led to FlepiMoP2 (2025/26). Supervising an undergraduate student in a scientific overhaul of the flu model for the 2026/27 round. • github.com/midas-network/flu-scenario-modeling-hub	2024 – present
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COVID-19 Scenario Modeling Hub ACCIDDA. Support. Data cleaning and assimilation in support of ACCIDDA's health heterogeneities COVID-19 round. • github.com/midas-network/covid19-scenario-modeling-hub	2025 – 2025
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Invited Talks and Seminars

FlepiMoP2 and the Operator-Partitioned Simulation Stack Tools Workshop, Insight Net Third Annual Meeting, Friday Center, Chapel Hill, North Carolina	Apr 2026
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Cross-Scale Feedback Motifs: Structure-Preserving Models and Computational Tools for Complex Systems Biological Oceanography Seminar, Woods Hole Oceanographic Institution, Woods Hole, Massachusetts	Mar 2026
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Decision-Support Modeling for One Health Pathogens Differential Equations Seminar, University of Maryland Baltimore County, Baltimore, Maryland	Mar 2026
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Modeling long-term viral infections Mathematics Seminar, Shippensburg University, Shippensburg, Pennsylvania	2025
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Scenario and forecast modeling for respiratory viruses Navajo Nation Epidemiology Center Data Summit, Window Rock, Arizona	2025
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Impacts of host behavior and viral life history on infectious disease transmission Department of Mathematics, Virginia Tech; Infectious Disease Dynamics Group, Johns Hopkins University	2024
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Within-host viral life histories predict FMDV transmission dynamics for African buffalo Hawlena Lab, Ben-Gurion University, Midreshet Ben Gurion, Israel	2024
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Modeling process-based effects of harmful algal blooms University of Essex (2024); Tel Aviv University (2023); Israel Oceanographic and Limnological Research, Haifa (2023); Inter-University Institute for Marine Sciences, Eilat (2023)	2023 – 2024
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Patterns of cultural variation in the Austronesian expansion Fortunato Lab, University of Oxford (virtual)	2023
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Consilience in FMDV ecology Gresham Lab, New York University; Obolski Lab, Tel Aviv University	2022
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Selected Conference Presentations

Decision-Support Modeling for One Health Pathogens: Using Mechanistic Models for Surveillance and Forecast Design Society for Mathematical Biology Annual Meeting, Graz, Austria; mini-symposium talk	2026
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Recovering Ecological Geometry: A Trait- and Depth-Structured IPDE Model of Plankton Dynamics	2025
Society for Mathematical Biology Annual Meeting, Edmonton, Alberta, Canada; mini-symposium talk	
Biologically Grounded Surrogate-Driven Parameter Inference for Sparsely Observed Dynamical Systems	2025
Symposium on Biomathematics and Ecology Education and Research (BEER), Fairfax, Virginia; contributed talk	
Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms	2023
Society for Mathematical Biology Annual Meeting, Columbus, Ohio; mini-symposium talk	
Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo	2022 – 2023
SIAM TXLA, Lafayette, Louisiana (2023); SMB Epidemiology-Immunology Midyear Meeting (2022, virtual)	
Dengue immunodynamics: antibody-dependent enhancement, cross-reactive immunity, and model-data confrontation	2022 – 2023
SIAM TXLA (2023); Mathematical Perspectives on Immuno-Biology, Bulgaria (2023); Center for Combatting Pandemics, Tel Aviv (2023); European Conference on Theoretical and Mathematical Biology, Heidelberg (2022)	
Disease dynamics necessarily operate across scales	2021
SIAM TXLA, South Padre Island, Texas; mini-symposium talk	
Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates	2021
Society for Mathematical Biology Annual Meeting (virtual); contributed talk	

Workshop Facilitation

Insight Net Third Annual Meeting , Tools Workshop Presenter	Chapel Hill, North Carolina, USA
Presenting FlepiMoP2 and the op_system/op_engine simulation stack at the Insight Net Third Annual Meeting tools workshop, Friday Center, University of North Carolina at Chapel Hill.	
	Apr 2026
Insight Net Second Tech Transfer Workshop , Technical Support	Chapel Hill, North Carolina, USA
ACCIDDA. Technical support for the Insight Net Second Tech Transfer Workshop at UNC Chapel Hill. Assisted participants with Python package installation and environment setup for Epydemics tools.	
	Jan 2026
ACCIDDA , Workshop Facilitator	Baltimore, Maryland, USA
Facilitator for the Applied Modeling in Public Health (AMPH) Workshop, October 15–17, 2025, Johns Hopkins Bloomberg School of Public Health.	
	Oct 2025
ACCIDDA , Workshop Facilitator	Baltimore, Maryland, USA
Facilitator for the Applied Modeling in Public Health (AMPH) Workshop, August 12–15, 2024, Johns Hopkins Bloomberg School of Public Health.	
	Aug 2024

Teaching

Johns Hopkins University , Teaching Assistant	Baltimore, Maryland, USA
Teaching assistant for <i>Topics in Infectious Disease Epidemiology</i> with Shaun Truelove and Sara Loo.	
	June 2025 – Aug 2025

University of Louisiana at Lafayette , Teaching Assistant and Instructor of Record Instructor of record for elementary statistics (3 sections), survey of calculus (2 sections), precalculus (2 sections), and college algebra (2 sections). Teaching assistant for calculus, linear algebra, differential equations, and basic statistics.	Lafayette, Louisiana, USA 2017 – 2020
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McDowell County Schools / Memorial University / UNC Asheville , Teaching and Tutoring Instructor of record for North Carolina Math III / Algebra II (2017). Mathematics and statistics tutoring at Memorial University of Newfoundland (2016–2017) and University of North Car- olina Asheville (2016).	2016 – 2017
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Mentoring

Mentoring students and postdoctoral researchers • Jolles and Beechler Labs, Oregon State University (2025–present) • Society for Mathematical Biology (2022–present) • Association for Women in Mathematics (2023–2025) • Ram Lab, Tel Aviv University (2022–2024) • MIDAS Network (2021–2022) • Gulbudak Group, University of Louisiana at Lafayette (2021–2022)	2021 – present
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Undergraduate Research Supervision: Shany Naim Tel Aviv University. Supervised undergraduate research project: initial translation of the VBPCPy variational Bayesian PCA codebase from MATLAB to Python.	2024
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Community Outreach

Science communication and research dissemination • University of Louisiana R1 Showcase: selected to present COVID-19 modeling research at the university-wide showcase (2022) • Ocean View United Methodist Church, Oak Island, NC: community discussion on vaccines and COVID-19, with on-site pharmacist for vaccination (2021)	2021 – 2022
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Professional Service

SMB Diversity, Equity, and Inclusion Committee Member	2023 – 2025
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Session Chair / Leader • SMB Early Career Workshop: Applying for postdoctoral funding (2023) • SMB Epidemiology-Immunology Midyear Meeting: Contributed talks (2022) • SMB Annual Meeting: CT04-MEPI contributed talks in mathematical epidemiology (2021)	2021 – 2023
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Mini-symposium Co-organizer • SIAM TXLA: Data-driven mathematical and statistical modeling approaches to population biology (2023) • SMB Annual Meeting: Disease Dynamics Across Scales (2023) • SMB/ESTMB Joint Meeting: Multi-scale and data-driven modeling approaches in ecology, immunology, and epidemiology (2022) • SIAM TXLA: Reproducibility, reliability, and robustness: confronting models from across mathematical biology with data (2021) • SIAM TXLA: Population Dynamics in Ecology and Epidemiology (2019)	2019 – 2023
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Secretary, University of Louisiana AMS Graduate Student Chapter	2021 – 2022
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Journal Reviewing

2022–present: *PLOS ONE*, *Journal of Biological Systems*, *Journal of Biological Dynamics*, *Frontiers in Public Health*, *Frontiers in Veterinary Science*, *Computational and Applied Mathematics*, *Mathematical Biosciences*, *Scientific Reports*, *PLOS Computational Biology*, and *BMC Infectious Diseases*.

Honors and Awards

Travel Awards

2019 – 2022

SIAM, Society for Mathematical Biology, and University of Louisiana Graduate Research Association

Rhodes Outstanding Graduate Assistant Award

2021

Professional Affiliations

Society for Mathematical Biology; Society for Industrial and Applied Mathematics; International Society for Bayesian Analysis; United States Research Software Engineer Association; MIDAS Network; Association for Women in Mathematics; Institute of Industrial and Systems Engineers; Society for Health Systems (SHS/IISE); and Healthcare Information and Management Systems Society (HIMSS).